

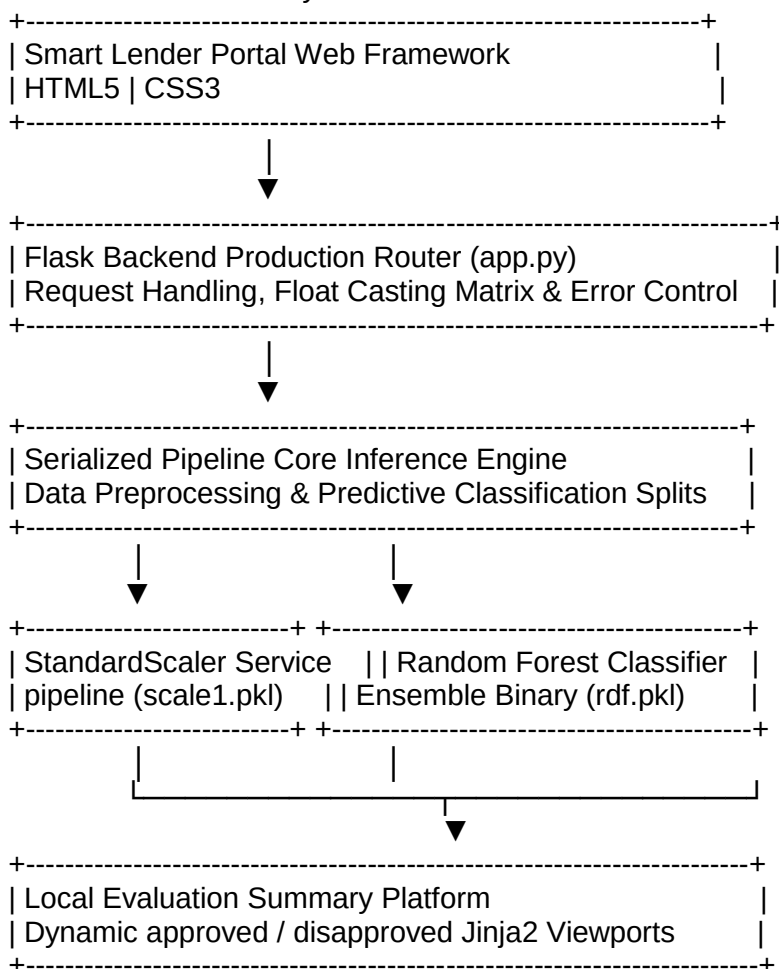
Solution Architecture

Date	02-July-2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	5 Marks

Solution Architecture Diagram

The diagram illustrates the flow from data acquisition to model deployment. Starting with the collection of loan applicant data, the process moves into a robust Data Engineering phase involving cleaning, balancing (SMOTE), and scaling. This ensures high-quality input for the core predictive engine: an optimized Random Forest classifier binary that generates automated risk scores. The entire inference pipeline is decoupled from the user interface and deployed as a multi-threaded Python/Flask web microservice to deliver real-time, low-latency eligibility determinations.

Presentation / Client Layer



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for entering applicant details and viewing loan eligibility results.	HTML5, CSS3,
API Gateway / Backend	Handles requests, validates input, and communicates with the ML model.	Python, Flask
Machine Learning Service	Analyzes applicant data and predicts loan eligibility.	Scikit-learn, XGBoost, Pandas, NumPy
External Integrations	Uses training datasets and optional notification services.	Kaggle Dataset, SMTP